Abu Dhabi Islamic Bank

Parallel approvals — five control functions, one decision, no queue

Wholesale credit approval routed through five independent control functions. Run in series it is a chain of queues; run in parallel it is a single elapsed wait — but only if dissent, dependency and amendment are designed for first.

5	1	3	1	0	100%
control functions, independent	elapsed wait instead of five	hard cases the design must handle	sixth function most banks lack	approvals re-sought on rework	decisions evidenced individually

APPROVAL SCOPE · WHOLESALE DEAL AND CREDIT LIFECYCLE

Business · Risk · Legal · Compliance · Sharia — each with independent authority and its own record

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Vendor, platform and system names removed. Approval structure is the client's; the design analysis, timing model and failure cases are mine.

EXECUTIVE SUMMARY

Serial approval makes elapsed time the sum of five queues; parallel makes it the longest one — and everything hard about the design lives in the exceptions

Situation

A wholesale credit facility requires approval from five control functions with genuinely independent authority: business, risk, legal, compliance and Sharia governance.

None of them reports to another. None can overrule another. Each owns a different question about the same facility.

Complication

Routed in series — the conventional design — the facility waits in five queues one after another, and total elapsed time is their sum. Each function also reviews a proposal that may already have been changed by the ones before it.

Parallel routing removes the queueing problem immediately. It also creates three problems that serial routing quietly avoided: what happens on dissent, what happens when one review genuinely depends on another, and what happens when the facility is amended after some approvals are already in.

The design

Route all five concurrently against a versioned facility. Treat dissent as a first-class outcome rather than a failure. Model the one genuine dependency explicitly instead of serialising everything to accommodate it.

And make amendment invalidate only the approvals whose subject actually changed — which is the single decision that determines whether parallel approval survives contact with real deals.

THE NUMBERS THAT MATTER

5	**$\Sigma \rightarrow$ max**	**3**	**1**	**1**	**5**
concurrent reviewers	elapsed time model	exception cases designed	genuine dependency	versioned subject	independent records

The judgement this argues: parallel approval is not a workflow optimisation. It is a claim that five functions can review the same artefact at the same time — which is only true if the artefact is versioned and amendment is scoped. Without those two things, parallel routing produces faster approvals of the wrong version.

HOW TO READ THIS DECK

The approval structure was the client's; the routing design, the exception handling and the timing model are mine

APPROVAL STRUCTURE

Client context

The control functions, their authority and their mandates as defined by the bank's governance framework.

- Five independent control functions
- Wholesale deal and credit lifecycle
- Delegated authority matrix
- Sharia governance mandate
- Audit and evidence obligations
- Existing systems of record

DELIVERY RECORD

My ownership

The parallel routing design, exception handling, evidence model and delivery to production.

- Parallel routing architecture
- Dissent and dependency handling
- Amendment invalidation scoping
- Per-function evidence model
- Control-function engagement
- Requirement through production

ILLUSTRATIVE

Modelled

Elapsed-time figures illustrate the serial-versus-parallel arithmetic. The ratio is the argument, not the values.

- Queue and review-time model
- Serial versus parallel comparison
- Amendment rework scenarios
- Replace with programme actuals
- Shape is the argument
- Method transfers unchanged

OUT OF SCOPE

Not claimed

Excluded deliberately. A design analysis, not a governance disclosure or a credit-process document.

- The bank's authority matrix
- Any credit decision or facility
- Any Sharia ruling or opinion
- Any confidential client document
- Measured cycle-time outcomes
- Portfolio or exposure data

Source: Author's delivery record (ownership) · author's design analysis (models and timing)

Each function asks a different question about the same facility — which is why none of them can answer for another, and why the order between them is arbitrary

Business

Should we do this deal at all?

- Commercial rationale and relationship fit
- Pricing against the risk taken
- Portfolio and concentration appetite
- Wallet and cross-sell implications

Risk

Can the counterparty repay?

- Credit assessment and internal rating
- Structure, security and covenants
- Limits, exposure and concentration
- Stress and downside scenarios

Legal

Is it enforceable as written?

- Documentation and contract form
- Security perfection and jurisdiction
- Counterparty capacity and authority
- Precedent and deviation from standard

Compliance

Are we permitted to do it?

- Sanctions, screening and know-your-customer
- Regulatory limits and reporting
- Conflicts and related-party rules
- Financial-crime risk

Sharia

Is the structure permissible?

- Contract form and its conformity
- Underlying asset and its treatment
- Prohibited activity screening
- Profit treatment and late-payment handling

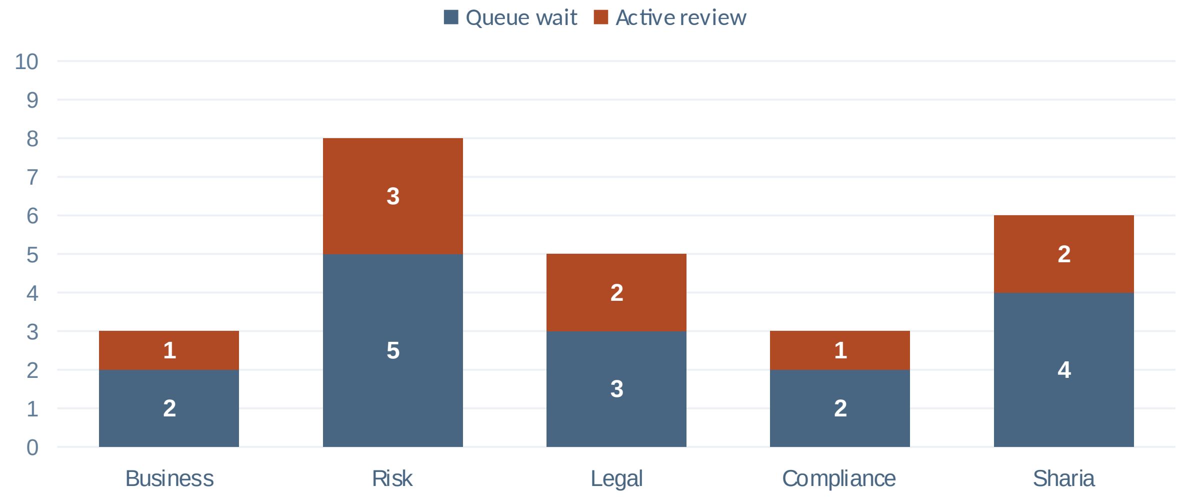
Why independence is the design constraint, not a governance detail

These are five separate mandates, not five stages of one review. No function can approve on another's behalf, none can overrule another, and a rejection from any one of them stops the facility regardless of what the other four concluded.

That has a direct architectural consequence. If the order between them is arbitrary — and it is, because none depends on another's conclusion — then serialising them is a pure cost with no governance benefit. The only reason approvals are usually serial is that a document physically moved between desks, and the workflow inherited the shape of the paper process rather than the shape of the governance.

Source: Author's delivery record, wholesale credit programme · author's analysis

Serial elapsed time is the sum of five queues; parallel is the longest one — and review effort is identical in both



Illustrative days per function — queue wait dominates active review in every case

25
days elapsed, routed in series

8
days elapsed, routed in parallel

9
review-days, identical either way

3×
reduction from routing alone

What the numbers show and what they hide

- Review effort is unchanged — nine reviewer-days either way, not a productivity claim.
- The entire gain comes from removing queues that existed only because of routing order.
- It hides the rework case — one unscoped amendment can erase the gain.

Why this framing matters when presenting to control functions

Approval-time initiatives usually arrive at control functions as an implied criticism — you are too slow. That framing guarantees resistance, and the functions are right to resist it, because their review time is not the problem.

Presenting the same change as removing queues that routing created, with reviewer effort explicitly unchanged, makes it a design conversation rather than a performance one. In practice this was the difference between five functions agreeing to a redesign and five functions defending their service levels.

Three cases serial routing quietly avoided — each has to be designed explicitly, and the third decides whether the model survives

01

Dissent

THE PROBLEM

One function rejects while others approve. In serial routing this stops the chain and the later functions never see the facility at all.

THE DESIGN

Treat rejection as a first-class outcome that halts progression but does not discard the approvals already given. Record every position, including those given before the rejection.

If dissent voids everything, the deal team resubmits from zero after each amendment — and parallel routing becomes slower than serial.

02

Genuine dependency

THE PROBLEM

One review really does need another's output. In this portfolio there was exactly one such case: the contract structure has to be settled before enforceability can be assessed.

THE DESIGN

Model that single dependency explicitly as a gate between two functions, and leave the other three unconstrained. Do not serialise all five to accommodate one pair.

The common failure is discovering one dependency and reverting the whole design to serial — which trades a real gain for a marginal one.

03

Amendment after partial approval

THE PROBLEM

The facility changes when three functions have approved and two have not. Every approval was given against a document that no longer exists.

THE DESIGN

Version the facility, and scope invalidation to the approvals whose subject actually changed. A pricing change invalidates business and risk; it does not invalidate enforceability or permissibility.

This is the decision the model lives or dies on. Blanket re-approval on any amendment restores the serial cost immediately, and amendments are not rare.

Scoped invalidation is the whole design. Everything else about parallel approval is straightforward once amendments stop resetting the board.

Sharia governance is a distinct authority reviewing a distinct object — modelling it as a variant of compliance is the most common design error

What it is not

- Not a compliance sub-check. Compliance asks whether the bank is permitted to act; Sharia governance asks whether the structure itself is permissible.
- Not advisory. It carries independent authority — a negative opinion stops the facility regardless of credit quality or commercial value.
- Not a screening rule. Prohibited-activity screening is one input; the substantive review is of the contract form and the underlying asset.
- Not resolvable by pricing. An impermissible structure is not a risk to be priced; it is ineligible.

What that requires of the design

- The contract structure has to exist as a first-class object on the facility, versioned and reviewable independently of the credit terms.
- The review must be able to conclude on the structure while credit terms are still moving — which is exactly what parallel routing enables.
- The opinion is a distinct record with its own identity, retention and audit trail, not a checkbox on a credit approval.
- Amendment scoping must know which changes touch the structure and which do not. A repricing does not reopen permissibility; a change of contract type does.

Why this is worth understanding beyond Islamic banking

The pattern generalises. Any market where an additional authority reviews an object that conventional systems treat as an attribute produces the same design problem — a sustainability mandate reviewing use of proceeds, a public-sector body reviewing procurement conformity, a co-operative structure reviewing member benefit.

In each case the platform question is identical: is this a new approver on the existing object, or a new object with its own approver? If the answer is the second and the platform models it as the first, the review has nowhere to attach, the evidence has no home, and the amendment logic cannot tell whether the review needs reopening.

That is a data-model decision made early and cheaply, or a retrofit made late and expensively. Recognising which one you are looking at is the transferable skill.

Five independent decisions need five independent records — a single approval status is not an audit trail, however convenient it is to report on

What is recorded	Why it has to be separate	Retained for	Owner
The decision itself	Each function's conclusion, independently attributable and not inferable from an aggregate status	Facility life plus policy	Function
The version reviewed	An approval is only meaningful against the document version it was given on	Facility life plus policy	Platform
The reviewer identity	Delegated authority is personal; the record must show who exercised it and under what limit	Facility life plus policy	Function
The stated basis	Conditions, caveats and reasoning — the part an auditor actually reads	Facility life plus policy	Function
Invalidation events	Which amendment reopened which approval, and why it did not reopen the others	Facility life plus policy	Platform
Dissent and resolution	A rejection and how it was addressed is more informative than an approval	Facility life plus policy	Platform

The shortcut to avoid

- Collapsing five decisions into one status field because reporting is easier that way.
- It works until an audit asks what legal concluded, on which version.
- After that the answer must be reconstructed from logs, if at all.

What the separation buys

- Every audit question is answered by a record, not a reconstruction.
- Amendment scoping becomes provable — you can show which approvals were reopened and which were not.
- Functions accept parallel routing when the record is visibly their own.

The reporting answer

- Aggregate status is derived from the five records, never stored instead of them.
- That keeps dashboards simple without making the audit trail lossy.
- A one-line decision that is almost impossible to retrofit.

WHAT TRANSFERS

Four rules for designing multi-authority approval

01 **Check whether the order is real**

Serial approval is usually a fossil of a paper process rather than a property of the governance. If no function depends on another's conclusion, the order is arbitrary and serialising is pure cost with no control benefit.

02 **Scope invalidation, or lose the gain**

Amendment after partial approval is the case that decides everything. Blanket re-approval on any change restores the serial cost immediately — and amendments are not the exception, they are the normal path.

03 **Model one dependency as one dependency**

Discovering that a single pair of reviews is genuinely ordered is not a reason to serialise all five. Gate the pair and leave the rest concurrent.

04 **Keep the records separate, derive the status**

Five independent authorities produce five independent records. Aggregate status is derived for reporting, never stored in their place — a decision that is trivial early and effectively impossible later.

Parallel approval is not a workflow optimisation. It is a claim that five functions can review the same artefact simultaneously — true only if the artefact is versioned and amendment is scoped, and false in an expensive way if it is not.

Source: Author's design analysis